

Yu Li

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Education

George Washington University <i>Ph.D. Candidate in Electrical and Computer Engineering, GPA: 4.0/4.0</i> Advisor: Prof. Tian Lan 📧; Work with: Prof. Zhengling Qi 📧	Washington, D.C. 2025.08 – 2029.05 (expected)
Wuhan University, Hongyi Honor College <i>B.Eng. in Microelectronics Science and Engineering, GPA: 3.87/4.0</i>	Wuhan, China 2021.09 – 2025.06

Experience

ByteDance, Tiktok <i>Research Scientist Intern, Recommendation Core</i>	San Jose, CA 2026.05 – 2026.08
Owned the ranking stage of TikTok video recommendation, working on 3B Large Recommendation Model with MLA, time-aware RoPE, and mixed-query optimization.	

Research Highlights

My research develops principled reinforcement learning formulations for **LLM Reasoning**, spanning token-level credit assignment, preference optimization, and context conditioning [NeurIPS'26] [NeurIPS'26][P2][P3], and extends these formulations to **Agentic RL** through hierarchical skill acquisition and multi-turn policy optimization [P1][ACL'26][INFOCOM'26]. On **Efficient AI**, my work focuses on parameter-efficient fine-tuning and long-context KV cache management [COLM'26][AAAI'26]. Earlier work on **Generative Models** covers diffusion-based personalized content-style synthesis [CVPR'26].

Preprints & Under Review

P=Preprint

- [P1] **ARISE: Agent Reasoning with Intrinsic Skill Evolution in Hierarchical Reinforcement Learning** 📧
[Yu Li](#), Rui Miao, Zhengling Qi, Tian Lan
- [P2] **Right Meets Wrong: Bilateral Context Conditioning with Reward-Confidence Correction for GRPO** 📧
[Yu Li](#), Tian Lan, Zhengling Qi
- [P3] **Hindsight Self-Distillation: Demonstrating to the Teacher for On-Policy LLM Reasoning** 📧
[Yu Li](#), Shu Hong, Tian Lan
Under review. ARR May 2026 OA: 3.33/5

Selected Publications [Full List]

†=Equal Contribution

- [NeurIPS'26] **OPPO: Bayesian Value Recursion for Token-Level Credit Assignment in LLM Reasoning**
[Yu Li](#), Rui Miao, Tian Lan, Zhengling Qi
- [NeurIPS'26] **INSPO: Unlocking Intrinsic Self-Reflection for LLM Preference Optimization**
[Yu Li](#), Tian Lan, Zhengling Qi
- [COLM'26] **MomentKV: Closing the Directional Gap in KV Cache Eviction for Long-Context Inference**
[Yu Li](#), Binxu Li, Tian Lan

[[ACL'26 Findings](#)] Reason in Chains, Learn in Trees: Self-Rectification and Grafting for Multi-turn Agent Policy Optimization

[Yu Li](#), Sizhe Tang, Tian Lan

[[CVPR'26](#)] CRAFT-LoRA: Content-Style Personalization via Rank-Constrained Adaptation and Training-Free Fusion

[Yu Li](#), Yujun Cai, Chi Zhang

[[ICASSP'26 Oral](#)] KG-SAM: Encoding Structural Constraints into Segment Anything Models via Probabilistic Graphical Models

[Yu Li](#), Da Chang, Xi Xiao

[[INFOCOM'26](#)] ACDZero: Graph-Embedding-Based Tree Search for Mastering Automated Cyber Defense

[Yu Li](#)[†], Sizhe Tang[†], Rongqian Chen, Fei Xu Yu, Guangyu Jiang, Mahdi Imani, Nathaniel D. Bastian, Tian Lan

[[ACM MM'26](#)] Prompted Information Bottlenecks: Rethinking Layer-Wise Information Allocation in Frozen Vision Transformers

Yuqi Li, Xi Xiao, Yunbei Zhang, Lin Zhao, [Yu Li](#), Aiden Zhao, Tianyang Wang, Hao Xu, Yingli Tian

[[AAAI'26](#)] Calibrating and Rotating: A Unified Framework for Weight Conditioning in PEFT

Da Chang, Peng Xue, [Yu Li](#), Yongxiang Liu, Pengxiang Xu, Shixun Zhang

Selected Awards & Honors

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- **ICML 2026 Gold Reviewer**, top reviewers recognition. 2026
 - **Innova International Exchange Scholarship**, 6 recipients university-wide, ¥70,000. 2024
 - **Innova Excellence Scholarship**, Top 3%, ¥10,000, twice. 2023, 2024
 - **First-Class Scholarship**, Top 5%, ¥3,000, three times. 2022, 2023, 2024

Academic Services

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- **Conference Reviewer**: 2027: AAAI. 2026: NeurIPS, ICML, ICLR, COLM, AAAI, ACM MM.
 - **Journal Reviewer**: TPAMI, TMLR, ToN, Neurocomputing.